Design and Analysis of Algorithms

Department of Computer Science & Engineering

DESIGN AND ANALYSIS OF ALGORITHMS

Dynamic Programming- Memory Function Knapsack

Department of Computer Science & Engineering

Bottom Up Approach

- Advantage of bottom up approach: each value computed only once

Memory Function Knapsack
Bottom Up Approach

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- Example computed bottom up:

	capacity *j*				
i	1	2	3	4	5
1	0	12	12	12	12
2	10	12	22	22	22
3	10	12	22	30	32
4	10	15	25	30	37

Memory Function Knapsack
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- Disadvantage of bottom up approach: values not required also computed

Memory Function Knapsack
Top Down Approach

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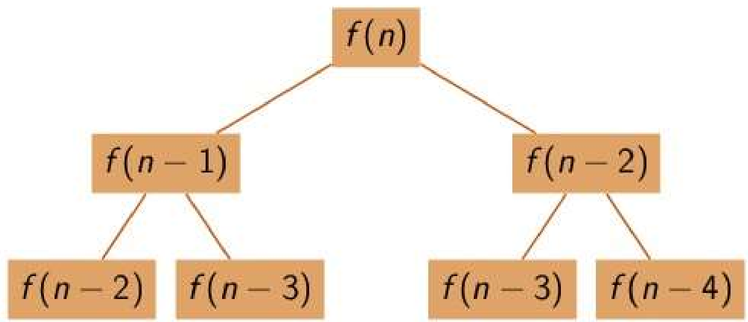
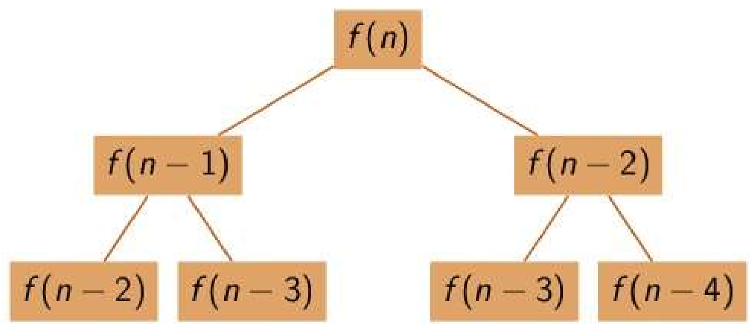
Memory Function Knapsack
Top Down Approach

- Disadvantage of top down approach: same problem solved multiple times
- Example computed top down:



Memory Function Knapsack
Top Down Approach

- Disadvantage of top down approach: same problem solved multiple times
- Example computed top down:



- Advantage of top down approach: only the required subproblems solved

Memory Function Dynamic Programming

- Combine the advantages of bottom up and top down approaches:
  - compute each subproblem only once
  - compute only the required subproblems

Memory Function Knapsack
MF-DP Algorithm

Algorithm for Memory Function Dynamic Programming

procedure MFKnapsack(i, j)
  ▷ **Inputs:** i indicating the number items, and
  ▷ j, indicating the knapsack capacity
  ▷ **Output:** The value of an optimal feasible subset of the first i items
  ▷ **Note:** Uses global variables input arrays $Weights[1 \dots n]$, $Values[1 \dots n]$,
  ▷ and table $F[0 \dots n, 0 \dots W]$ whose entries are initialized with −1's except
  ▷ row 0 and column 0 is initialized with 0
  if $F[i, j] < 0$ **then**
    if $j < Weights[i]$ **then**
      $value \leftarrow MFKnapsack(i-1, j)$
    else $value \leftarrow max(MFKnapsack(i-1, j),\ Values[i] + MFKnapsack(i-1, j - Weights[i]))$
      $F[i, j] \leftarrow value$
  return $F[i, j]$

Memory Function Knapsack
Example

$$F(i,j) = \begin{cases} \max(F(i-1,j),\ \ v_i + F(i-1,j-w_i)) & \text{if } j - w_i \geq 0 \\ F(i-1,j) & \text{if } j - w_i < 0 \end{cases}$$

Dynamic Programming Example

item i	weight w_i	value v_i
1	2	12
2	1	10
3	3	20
4	2	15

What is the maximum value that can be stored in a knapsack of capacity 5?

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0	0					
1						
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4						$\square$

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2	□	–	□	□	–	□
3	□	–	–	□	–	□
4	□	–	–	–	–	□

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Knapsack problem solved by

- computing 21 out 30 possible subproblems
- reusing subproblem entry $(1,2)$

Memory Function Knapsack

Complexity

- Constant factor imrpovement in efficiency
  - Space complexity: $\Theta(nW)$
  - Time complexity: $\Theta(nW)$
  - Time to compose optimal solution: $O(n)$
- Bigger gains possible where computation of a subproblem takes more than constant time

Think About It

- Consider the use of the MF technique to compute binomial coefficient using the recurrence

$$C(n,k) = C(n-1,k-1) + C(n-1,k)$$

  - How many table entires are filled?
  - How many are reused?